

Sujatro Ganguli

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PROFESSIONAL SUMMARY

Third Year Computer Science student proficient in Python, AI Engineering, and Web Development. Expert in RAG frameworks and open-source libraries. Proven leader, as Technical Head at SEDS VIT, delivering high-availability platforms and automated research software.

EDUCATION

Vellore Institute of Technology (VIT), Vellore — B.Tech in CSE (CGPA: 8.64) 2023 – Present

TECHNICAL SKILLS

Languages: Python (Expert), C++, C, Java, JavaScript, TypeScript, SQL, Quickshell

Frameworks & Tools: React, Next.js, Bun, Vercel, Railway, FastAPI, Streamlit, Git, Cloudflare, OpenCV, Wayland

Domains: RAG Systems, Full-Stack Development, ML Optimization, API Design, Linux Systems

EXPERIENCE

- **Technical Head — SEDS VIT:** Architected event portals for Hermes 🌐 and Erebus 🌐; managed **270+ registrations** with **100% uptime** via Vercel and Cloudflare DNS. 2026
 - **Research Intern — PRL:** Automated SHARAD radar visualization using Python; reduced rendering latency for **3 GB of data** from hours to **sub-10 seconds**. 2025
 - **Web Volunteer — Outreach:** Redesigned websites for two NGOs using HTML/CSS; improved accessibility and dedicated **100+ hours** to community initiatives with NGO CINI. 2021 – 2023
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TECHNICAL PROJECTS

- **HyprQuickPaper:** Wayland wallpaper manager optimized for sub-200ms switching and 0% idle CPU usage. *Tech: Quickshell, Wayland, Shell, Linux* 🐙
- **EVA — RAG Tool:** Engineered .eva format for versioned indexing, reducing re-indexing time from hours to seconds. *Tech: Python, RAG, Vector Indexing, LLM* 🐙
- **Gestualis:** Low-power gesture library using dot-product logic for 100% consistency. *Tech: Python, OpenCV, Computer Vision, MediaPipe* 🐙
- **Automated Question Gen:** RAG pipeline automating academic assessment creation, reducing manual effort by 75%. *Tech: Python, RAG, NLP, AI Engineering* 🐙
- **Exoskeleton Optimization:** Applied NSGA-II via MATLAB to optimize biomechanical parameters in real-time. *Tech: MATLAB, NSGA-II, Optimization Algorithms* 🐙
- **Hermes Portal:** Architected event portal for SEDS VIT managing 270+ registrations with 100% uptime. *Tech: React, Next.js, Vercel, Cloudflare DNS* 🌐
- **Erebus Portal:** High-availability registration system for SEDS VIT events with integrated DNS management. *Tech: React, Next.js, Vercel, Cloudflare DNS* 🌐
- **OriginShift:** Algorithm designed to generate randomized yet guaranteed solvable mazes. *Tech: Python, Algorithms, Maze Generation, Data Structures* 🐙
- **HackBattle (Tracker):** Hackathon project that tracks hand and eye movement to control mouse pointer navigation. *Tech: JavaScript, Computer Vision, Webcam API, Tracking* 🌐

- **HyprTodo:** Todo management system utilizing shell scripts for task execution and Quickshell for the interface. *Tech: Shell Scripting, Quickshell, Wayland, Linux* 🐧
 - **Sentimental Text:** Modular CLI tool for sentiment analysis featuring preprocessing, feature extraction, and deep learning models. *Tech: Python, Deep Learning, NLP, CLI* 🐧
 - **Arabella:** AI-powered search tool utilizing a Graph Database to store and retrieve semantic connections. *Tech: Python, KuzuDB, GraphDB, Search Engine, AI* 🐧
 - **DSW (Room Booking):** Management system for handling room bookings and scheduling. *Tech: Web Development, Database Management, HTML/CSS* 🐧
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AWARDS & ACHIEVEMENTS

- **1st Place — Caterpillar Hackathon (2024):** Built technician app with face-ID and voice-to-text.
 - **Track Prize — Yantra Central Hackathon (2025):** Developed genomic analysis platform with **89% accuracy**.
 - **1st Place — Prima Vera Filmmaking (2023) — National Qualifier — First Lego League (2019)**
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CERTIFICATIONS & INTERESTS

Certifications: DELF B1 (French), Udemy Python Programming (64.5 hours)

Languages: English (Professional), French (Conversational), Hindi, Bengali

Interests: Linux System Customization, Machine Learning Research, Theatre